

Data Structure

Problem Statement: Student Attendance Management System

Design a simple system to manage student attendance using basic data structures.

Requirements:

1. Store student IDs in the order they enter the classroom.
2. Quickly check whether a student is present using their ID.
3. Allow the last marked attendance to be undone.
4. Display all present students in the order they arrived.

Constraints:

- Student IDs are unique.
- Attendance is marked by one student at a time.

Data Structures to Use:

- **Queue** → Maintain entry order
- **Hash Set / Hash Map** → Fast presence check
- **Stack** → Undo last attendance
- **Array / List** → Display all records

Expected Outcome:

The system should efficiently add, search, undo, and display student attendance.